

Test 2 Set W

Physics

Date: August 30, 2026

Full Marks: 100

Class – XI

Time: 50.0 Minutes

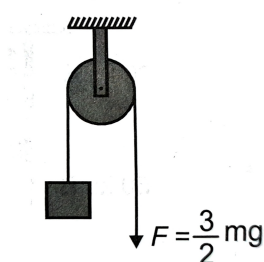
General Instructions

1. **Negetive Marking** is -1 for a single answer wrong
2. Use **blue** or **black** ink only.
3. Mobile phones, calculators, and other electronic devices are not permitted.
4. There is only Section A but in your OMR fill the questions 21 to 25 under Section B.

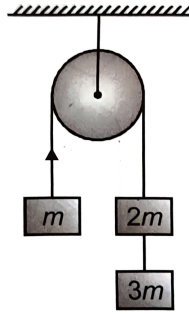
SECTION A: Single Correct Answer Type

1. A rocket of mass 5700 kg ejects mass at a constant rate of 15 kg/s with constant speed of 12 km/s. The acceleration of the rocket 1 minute after the blast is ($g = 10 \text{ m/s}^2$)
(A) 34.9 m/s^2 (B) 3.50 m/s^2
(C) 27.5 m/s^2 (D) 13.5 m/s^2
2. An unloaded bus can be stopped by applying brakes on straight road after covering a distance x . Suppose, the passenger add 50% of its weight as the load and braking force remains unchanged, how far will the bus go after the application of the brakes? (Velocity of bus in both case is same)
(A) 0 (B) $2.5x$
(C) $1.5x$ (D) $2x$
3. A spring of force constant K is first stretched by distance a from its natural length and then further by distance b . The work done in stretching the part b is
(A) $\frac{1}{2}Kb(a - b)$
(B) $\frac{1}{2}Kb(2a + b)$
(C) $\frac{1}{2}Ka(a - b)$
(D) $\frac{1}{2}Ka(a + b)$
4. A rifle bullet loses $\frac{1}{20}$ th of its velocity in passing through a plank. Assuming that the plank exerts a constant retarding force, the least number of such planks required just to stop the bullet is
(A) Infinite (B) 11
(C) 21 (D) 20
5. Newton's first law is applicable
(A) Only in non-inertial reference frames
(B) In all reference frames
(C) Only in inertial reference frames
(D) None of these

6. KE acquired by a mass m in travelling a certain distance d , starting from rest, under the action of a constant force F is
- (A) Directly proportional to $\sqrt{m}$
 (B) Directly proportional to $\frac{1}{m}$
 (C) Directly proportional to m
 (D) None of these
7. A U-238 nucleus, originally at rest, decays by emitting an α -particle, say with a velocity of v m/s. The recoil velocity (in ms^{-1}) of the residual nucleus is
- (A) $-\frac{4v}{238}$ (B) $-\frac{4v}{238}$
 (C) $\frac{4v}{238}$ (D) $\frac{v}{4}$
8. A block of mass m is at rest on a rough inclined plane of angle of inclination θ . If coefficient of friction between the block and the inclined plane is μ , then the minimum value of force along the plane required to move the block on the plane is
- (A) $mg[\sin \theta - \mu \cos \theta]$
 (B) $mg[\sin \theta + \mu \cos \theta]$
 (C) $mg[\mu \cos \theta - \sin \theta]$
 (D) $mg[\mu \cos \theta + \sin \theta]$
9. A machine gun fires a bullet of mass 65 g with a velocity of 1300 m/s. The man holding it can exert a maximum force of 169 N on the gun. The number of bullets he can fire per second will be
- (A) 4 (B) 2
 (C) 3 (D) 1
10. In the arrangement shown, the mass m will ascend with an acceleration (Pulley and rope are massless)



- (A) $2g$ (B) $\frac{g}{2}$
 (C) Zero (D) g
11. A spring with spring constant k when compressed by 1 cm, the potential energy stored is U . If it is further compressed by 3 cm, then change in its potential energy
- (A) $9U$ (B) $15U$
 (C) $3U$ (D) $8U$
12. A string tied on a roof can bear a maximum tension of 50 kg wt. The minimum acceleration that can be acquired by a man of 98 kg to descend will be [Take $g = 9.8 \text{ m/s}^2$]
- (A) 9.8 m/s^2
 (B) 5 m/s^2
 (C) 4.9 m/s^2
 (D) 4.8 m/s^2
13. A uniform chain of length L and mass M is lying on a smooth table and one third of its length is hanging vertically down over the edge of the table. If g is acceleration due to gravity, the minimum work required to pull the hanging part of the chain on the table is
- (A) $\frac{MgL}{3}$ (B) $\frac{MgL}{9}$
 (C) $\frac{MgL}{18}$ (D) MgL
14. A stationary particle explodes into two particles of masses x and y , which move in opposite directions with velocity v_1 and v_2 . The ratio of their kinetic energies ($E_1 : E_2$) is
- (A) $\frac{x}{y}$ (B) 1
 (C) $\frac{y}{x}$ (D) $\frac{xv_2}{yv_1}$
15. In the figure given below, with what acceleration does the block of mass m move? (Pulley and strings are massless and frictionless)

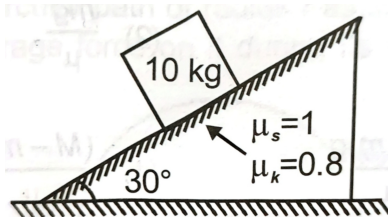


- (A) $\frac{2g}{5}$ (B) $\frac{2g}{3}$
(C) $\frac{g}{3}$ (D) $\frac{g}{2}$

16. In accordance with Newton's third law of motion

- (A) For appearance of action and reaction, physical contact is not necessary
(B) This law is applicable whether the bodies are at rest or they are in motion
(C) All of these
(D) Action and reaction never balance each other

17. A block of 10 kg mass is placed on a rough inclined plane as shown in figure. The acceleration of the block will be



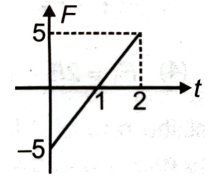
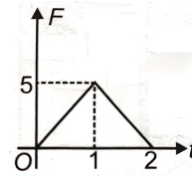
- (A) g (B) Zero
(C) $\frac{g}{2}$ (D) $\frac{\sqrt{3}g}{2}$

18. The power of water pump is 4 kW. If $g = 10 \text{ m s}^{-2}$, the amount of water it can raise in 1 minute to a height of 20 m is

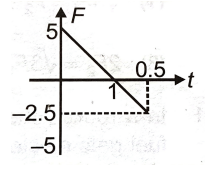
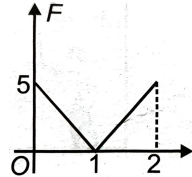
- (A) 1200 litre (B) 100 litre
(C) 2000 litre (D) 1000 litre

19. In which of the following graphs, the total change in momentum is zero?

- (A) (B)



- (C) (D)



20. If a force of constant magnitude acts in direction perpendicular to the motion of a particle, then its

- (A) Momentum is uniform
(B) All of these
(C) Speed is uniform
(D) Velocity is uniform

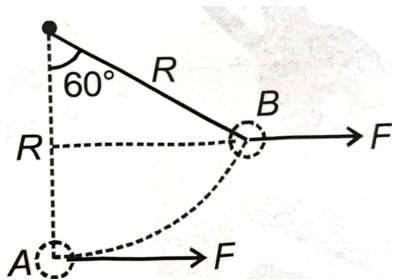
21. A 10 g bullet moving at 200 m/s stops after penetrating 5 cm of wooden plank. The average force exerted on the bullet will be

- (A) 4000 N
(B) 2000 N
(C) -4000 N
(D) -2000 N

22. The position x of a particle moving along x-axis at time t is given by the equation $t = \sqrt{x} + 2$, where x is in metres and t in seconds. Find the work done by the force in first four seconds.

- (A) 8 J (B) 4 J
(C) Zero (D) 2 J

23. A block of mass m is pulled along a circular arc by means of a constant horizontal force F as shown. Work done by this force in pulling the block from A to B is



- (A) FR (B) $\frac{\sqrt{3}}{2}FR$
 (C) mgR (D) $\frac{FR}{2}$

24. A particle of mass $2m$ moving with velocity v strikes a stationary particle of mass $3m$ and sticks to it. The speed of the system will be

- (A) $0.8v$ (B) $0.4v$
 (C) $0.2v$ (D) $0.6v$

25. A ball of mass m moving with velocity u collides head-on with the second ball of mass m at rest. If the coefficient of restitution is e and velocity of first ball after collision is v_1 and velocity of second ball after collision is v_2 then

- (A) $v_1 = \frac{(1-e)u}{2}$, $v_2 = \frac{(1+e)u}{2}$
 (B) $v_1 = \frac{u}{2}$, $v_2 = -\frac{u}{2}$
 (C) $v_1 = \frac{(1+e)u}{2}$, $v_2 = \frac{(1-e)u}{2}$
 (D) $v_1 = (1+e)u$, $v_2 = (1-e)u$

Answer Key

- | | | | | |
|--------|---------|---------|---------|---------|
| 1. (C) | 7. (A) | 13. (C) | 19. (B) | 25. (A) |
| 2. (C) | 8. (C) | 14. (C) | 20. (C) | |
| 3. (B) | 9. (B) | 15. (B) | 21. (C) | |
| 4. (B) | 10. (B) | 16. (C) | 22. (C) | |
| 5. (C) | 11. (B) | 17. (B) | 23. (B) | |
| 6. (D) | 12. (D) | 18. (A) | 24. (B) | |